

Statement of Work and Specifications for a high-precision 8" polishing system.

The system shall have the ability to eliminate user variability through high-precision, semiautomatic controls specifically designed for delicate microscopic sample preparation.

The system shall have precision "pitch and roll" micrometers that allow for angular positioning adjustments in fine increments.

The system shall provide 1 micron resolution for tracking material removal in real time.

The system shall provide users with the ability to preset specific removal amounts, allowing for unattended operation once target depth is reached.

The system shall have a drive specifically designed for delicate, unencapsulated TEM wedge thinning to absorb subtle movements that might otherwise fracture the sample.

The system shall allow for the quick removal of sample fixtures for inspection and exact repositioning afterward, ensuring the polishing plane remains the same.

The system shall provide independent control over both sample oscillation and rotation, which can be full 360° or limited back-and-forth "sweep" motions to simulate hand-polishing and reduce stress on fragile materials.

The system shall have a dual camera system that allows for precise sample alignment during testing along with a secondary camera for photo-documentation.

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